

Backpropagation explained

$$\frac{\partial L}{\partial z^{[2]}} = \delta^{[2]} = \underline{a^{[2]}} - y \quad (15)$$

$$\frac{\partial L}{\partial w^{[2]}} = \sum \underline{[a^{[1]}]^T}$$

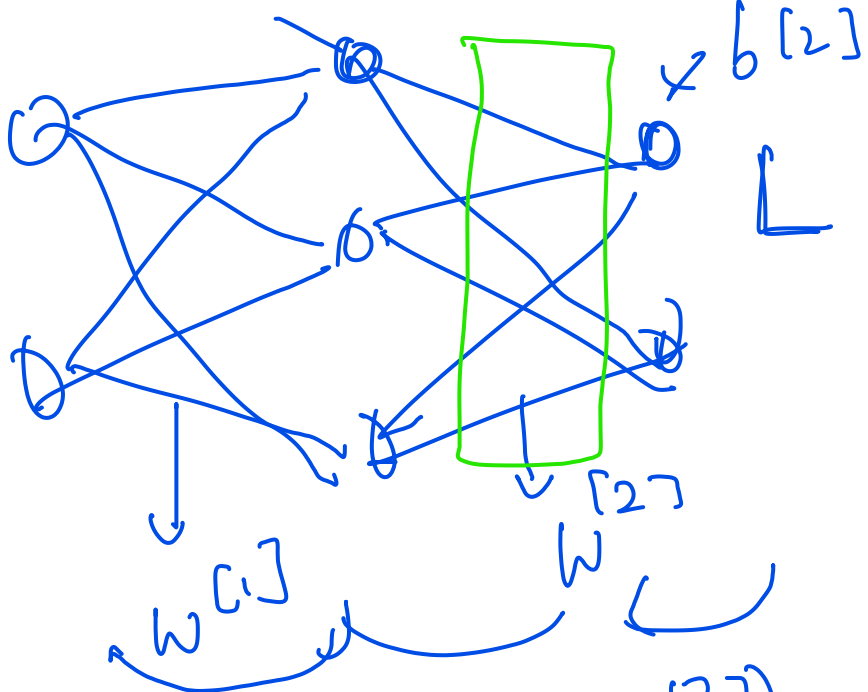
$$z^{[2]} \xrightarrow{\sigma} a^{[2]} \rightarrow L$$

$$w^{[2]_{\text{new}}} = w^{[2]_{\text{old}}} - \eta \left[\frac{\partial L}{\partial w^{[2]}} \right]$$

$$\underline{z^{[2]}} = \sigma(\underline{z^{[2]}})$$

$$L = -[y \log(\hat{y}) + (1-y) \log(1-\hat{y})]$$

$$\underline{L} = -[y \log(\underline{a^{[2]}}) + (1-y) \log(1-\underline{a^{[2]}})]$$



L is dependent on $a^{[2]}$

↓

$z^{[2]}$

↓

$$L = - [\gamma \log(a^{[2]}) + (1-\gamma) \log(1-a^{[2]})]$$

$$\frac{\partial L}{\partial z^{[2]}} = \left(\frac{\partial L}{\partial a^{[2]}} \right) \cdot \left(\frac{\partial a^{[2]}}{\partial z^{[2]}} \right) \quad (*)$$

$$\frac{\partial L}{\partial a^{[2]}} = \left[\gamma \frac{1}{a^{[2]}} + (1-\gamma) \frac{1}{1-a^{[2]}} \right]$$

$$= -\frac{\gamma}{a^{[2]}} + \frac{(1-\gamma)}{1-a^{[2]}}$$

$$= \frac{a^{[2]} - \gamma}{a^{[2]}(1-a^{[2]})} \times \cancel{a^{[2]}(1-a^{[2]})}$$

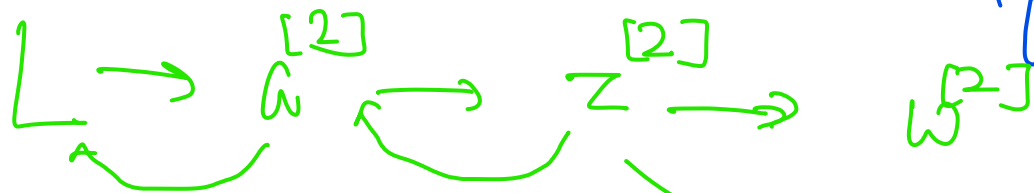
$$\frac{d}{dn} (\log(n^2)) = \frac{1 \times 2n}{n^2} = \frac{-\gamma(1-a^{[2]}) + (1-\gamma)a^{[2]}}{a^{[2]}(1-a^{[2]})} = \boxed{\frac{a^{[2]} - \gamma}{a^{[2]}(1-a^{[2]})}} \quad \frac{d}{dn} (1-n) = \frac{1(-1)}{1-n}$$

$$\frac{d}{dn} (\log(1-n)) = \frac{1(0-1)}{1-n} = \frac{-1}{1-n}$$

$$a^{[2]} = \sigma(z^{[2]})$$

↓
Sigmoid

$$\frac{\partial a^{[2]}}{\partial z^{[2]}} = \sigma(z^{[2]}) (1 - \sigma(z^{[2]})) = \boxed{a^{[2]}(1-a^{[2]})}$$



$$\frac{\partial L}{\partial w^{[2]}} = \left(\frac{\partial L}{\partial a^{[2]}} \right) \times \left(\frac{\partial a^{[2]}}{\partial z^{[2]}} \right) \times \left(\frac{\partial z^{[2]}}{\partial w^{[2]}} \right) = \boxed{a^{[1]} \times w^{[2]} + b^{[2]}}$$

$$\frac{\partial z^{[2]}}{\partial b^2} = 0 + 1$$

$$\frac{a^{[2]} - y}{a^{[2]}(1 - a^{[2]})} \times \cancel{a^{[2]}(1 - a^{[2]})} \times a^{[1]T} = \underbrace{(a^{[2]} - y)}_{\delta^{[2]}} a^{[1]T}$$

$$\left[\frac{\partial L}{\partial b^{[2]}} \right] = \frac{\partial L}{\partial a^{[2]}} \times \frac{\partial a^{[2]}}{\partial z^{[2]}} \cdot \left(\frac{\partial z^{[2]}}{\partial b^{[2]}} \right)$$

$$= \frac{a^{[2]} - y}{a^{[2]}(1 - a^{[2]})} \times \cancel{a^{[2]}(1 - a^{[2]})} \cdot 1 = (a^{[2]} - y) = \delta^{[2]}$$

Hidden layer $L \rightarrow a^{[2]} \rightarrow z^{[2]} \rightarrow w^{[2]} \rightarrow [F_{w'}] \times a^1 \rightarrow z^{[1]} \rightarrow w^1$



gradient of activation function
at value 1

$$z^{[1]} = \underline{\underline{\text{input} \times w^{[1]} + b^{[1]}}}$$

$$\text{Relu max}(0, x)$$

=

$$g(x)$$

$$\begin{bmatrix} x & > 0 \\ 0 & < 0 \\ x & > 0 \end{bmatrix}$$

$$g'(x) = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

$\frac{\partial L}{\partial w^1}$

vanishing

$$w'^{\text{new}} = w^{\text{old}} - \eta \frac{\partial L}{\partial w^1}$$

$m = \text{gradient signal}$

$$= 0.2$$

$=$

$$= \begin{bmatrix} 0.2 & 0.2 \\ 1 & 1.8 \end{bmatrix} - 0.1 \underbrace{\begin{bmatrix} 0 & \dots & 2 & 0 & \dots & 3 \\ 0 & \dots & 2 & 0 & \dots & 3 \end{bmatrix}}_{\approx 0}$$

$$g_{\text{out}} = 0.8$$

$$L = 10$$

$$\underline{\underline{g_{\text{int}}}} = \underline{g_{\text{out}}} \times m$$

$$m = 1.8$$

$$= 0.8 \times (0.25)^{10}$$

$$= 0.8 \times 0.0000000001$$

$0 < m < 1$
various

$$f_{\text{int}} = f_{\text{out}} \times m$$

$$= 0.8 \times (0.25)^{10}$$

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$m > 1$ Exponential

